

A designed null at every scale: spatial-scale and autocorrelation-robust sensitivity of a satellite-NO₂ steel-activity signal for the Tangshan cluster

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Abstract

The NOXUS pipeline turns Sentinel-5P/TROPOMI tropospheric NO₂ over the Tangshan (Hebei) steel cluster into a relative steel-activity signal and tests it against the CREA blast-furnace (BF) operating-rate benchmark. Earlier phases reported an apparent *decoupling*: on a partial NO₂ cube the meteorology-normalised cluster signal correlated negatively with the BF rate in levels (monthly $r = -0.35$) and only weakly after detrending. Two suspicions remained: that the result was an artefact of a single fixed *spatial scale*, and that the significance was inflated by treating strongly *autocorrelated* monthly series as i.i.d. This paper (NOX-008) resolves both on the complete, rebuilt 389-week cube (2019–2026). We sweep two scale axes by aggregation only — area-of-interest (AOI) extent (0.25° vs 0.10° buffer) and grid resolution (native $\sim 0.04^\circ$, block-averaged $\sim 0.1^\circ$ and $\sim 0.25^\circ$) — and re-judge every correlation with an autocorrelation-aware test: first-order effective sample size, a moving-block bootstrap confidence interval, a block-permutation null, and a Benjamini–Hochberg multiple-testing correction across the whole grid. The result is an emphatic, designed-for null: across 16 scale $\times$ variant cells, **zero** are significant even at the naive $p < 0.05$ threshold and **zero** survive the autocorrelation-robust test; the sign of the levels relationship is itself unstable across scale (–, +, –, –); and the coarsest grid cannot resolve a source/background contrast at all. The full series collapses the partial-cube -0.35 to -0.05 . We conclude that the weak/null NO₂ $\leftrightarrow$ BF coupling is not an artefact of one scale or of an over-optimistic significance test: it is robust to both. This hardens the project conclusion from “no signal at one scale” to “no robust signal at any reasonable scale,” and motivates the two remaining levers — a diurnal geostationary sensor (GEMS) and flux-divergence point-source attribution.

1 Background and antecedents

The premise. Satellite NO₂ is an established proxy for combustion activity, and a small literature uses it as an economic nowcast: at national scale for China (Morris and Zhang, 2019; Li and Zheng, 2023; Li et al., 2024), across global cities (Montgomery and Holloway, 2018), and as a war/industrial-shock read for Ukraine (Mao et al., 2025); the broad case for satellite data in economics is made by Donaldson and Storeygard (2016) and Ezran et al. (2023). NOXUS asks a deliberately harder, single-cluster question: can the NO₂ column over one industrial cluster — the Tangshan blast-furnace complex, the largest steel agglomeration in China — track that cluster’s physical output at a useful frequency, using only public, free data?

The NOXUS arc. Prior phases built the machinery and tested it honestly: NOX-001 ingested the CREA Tangshan BF operating-rate benchmark (CREA, 2026); NOX-002 acquired TROPOMI

NO₂ over the AOI via openEO and composited it into an analysis-ready weekly cube; NOX-003 derived a relative steel-activity signal using the cluster/relative-index paradigm (Kim et al., 2023), with the dominant ventilation confounder regressed out via ERA5 winds and boundary-layer height (Hersbach et al., 2020; C3S CDS, 2026; Li et al., 2024). NOX-003 returned an *honest near-null*: in levels the cluster signal correlated *negatively* with the BF rate (monthly ~ -0.35), the fingerprint of *decoupling* — emission intensity (NO_x per tonne) falling secularly faster than activity rises, driven by ultra-low-emission retrofits and policy (Li et al., 2024). NOX-003.1 modelled that intensity trend explicitly and confirmed only a weak positive residual relationship ($r \sim 0.14$, below the pre-registered 0.5–0.75 bar). NOX-004 reframed the signal as a discrete *event* marker rather than a continuous tracker, and NOX-006 added a daily Prophet decomposition to exploit the weekly cycle for source separation — which TROPOMI’s single $\sim 13:30$ overpass renders invisible (weekly term flat, 0.5% of variance). A decisive full-series *arbiter* re-ran the chain on the complete, rebuilt 389-week cube and found the average coupling collapses (weekly $r = 0.07$, monthly $r = 0.13$), with a regime-dependent rolling peak. The full series does not rescue the signal.

Two unresolved suspicions. Every prior number was computed at a *single* spatial scale — one AOI buffer, one native grid — and judged with the `scipy` Pearson p -value, which assumes independent observations. Both choices can manufacture or destroy a correlation:

1. **Scale.** Parubets and Naito (2025) show, for Japanese local economic indicators, that an NO₂↔activity correlation that is significant at a coarse spatial scale can weaken, vanish, or flip sign at a finer one. Steel is only ~ 30 –43% of Tangshan’s NO₂ column (Sci. Rep., 2024); a tighter AOI might isolate the source from the Beijing–Tianjin–Hebei background, or a coarser grid might average the cluster into that background.
2. **Serial dependence.** NO₂ and the BF rate are smooth, strongly autocorrelated series. The effective number of independent observations is far below the nominal n , so a naive p -value and the $\pm 1.96/\sqrt{n}$ lead-lag band overstate significance (Morris and Zhang, 2019).

This paper. NOX-008 confronts both suspicions on the complete cube. We sweep extent and resolution by *aggregation only* (never interpolation to a finer grid) and re-judge every correlation with an autocorrelation-robust test and a multiple-testing correction across the whole scale grid. An honest null is an acceptable, designed-for outcome (Morris and Zhang, 2019); the contribution is to show *how robust* that null is.

2 Data

Tropospheric NO₂ cube. The analysis-ready product is the NOX-002b weekly composite cube (`no2_cube_w.nc`): 30×51 cells at native $\sim 0.035^\circ \times 0.055^\circ$ (~ 4 –6 km), 389 weekly slots spanning 2019-01-06 to 2026-06-14, with per-cell valid-coverage tracked and cloud-gapped cells left as `NaN` (never interpolated). The AOI is *derived* from the located steel facilities (GEM Global Iron & Steel Tracker): the bounding envelope of all TROPOMI-era integrated BF/BOF plants in Tangshan prefecture (117.964–119.047°E, 38.954–40.210°N), expanded by a configurable buffer.

Physical-output benchmark. The CREA Tangshan blast-furnace operating-rate series (CREA, 2026) — weekly, WIND-sourced, republished freely by CREA — is the ground truth. It rises secularly across the window (Fig. 3b, blue), the hallmark of recovering and consolidating output.

Meteorology. ERA5 single-levels 10 m wind (u_{10}, v_{10}) and boundary-layer height (**blh**) at the overpass window (Hersbach et al., 2020; C3S CDS, 2026), aggregated over the footprint and regressed out of the cluster signal as the dominant ventilation confounder (Li et al., 2024). The meteorological covariate is regional and *scale-independent*, so the identical covariate set is applied at every spatial scale.

3 Methods

3.1 Signal chain (canonical, per scale)

For each spatial scale we reproduce the NOX-003 chain exactly. (i) **Footprint–background contrast**: select cube cells within a haversine radius of any operating facility (the footprint), and an annular background ring around the cluster centroid clipped to the AOI; the signal is the per-week footprint mean minus the background mean. (ii) **Meteo regress-out**: OLS-residualise the footprint signal on the ERA5 covariates $\{u_{10}, v_{10}, \text{blh}, \text{wind speed}\}$. (iii) **Deseason variants**: from the meteo-free monthly series we build four representations — **level** (no detrending), **intensity** (explicit smoothness-controlled secular-trend residual, the NOX-003.1 decoupling model), **yoy** (year-over-year difference), and **stl** (STL residual). Each is aligned to the monthly BF rate and correlated at lag 0 and at its peak- $|r|$ lead-lag (± 8 months).

3.2 Two scale axes (aggregation only)

Axis 1 — AOI extent. A strict spatial clip of the cube to the facility envelope plus a buffer: wide 0.25° (~ 28 km, the default) vs **tight** 0.10° (~ 11 km). The tight clip tests the source-isolation intuition: less external background, but also fewer cells. **Axis 2 — grid resolution.** Block-averaging the cube by integer factors over (y, x) via `xarray.coarsen` (mean over cell blocks): **native** (1,1), **coarse010** (3,2) $\approx 0.1^\circ$, and **coarse025** (7,5) $\approx 0.25^\circ$. Coarsening by averaging is the permitted, information-preserving direction (consistent with the project’s native-resolution decision, which rejected interpolating to a *finer* grid); it is exactly the Parubets and Naito (2025) resolution experiment. The footprint radius and ring floor at the native config (15 km; 25–60 km) and scale with the cell size so coarse grids still resolve a non-empty, disjoint geometry where possible.

3.3 Autocorrelation-robust significance

For every cell we report, alongside the naive Pearson r and p :

- **Effective sample size** $n_{\text{eff}} = n(1 - ab)/(1 + ab)$, with a, b the lag-1 autocorrelations of the two series (Bayley–Hammersley first order), and the p -value recomputed on $n_{\text{eff}} - 2$ degrees of freedom.
- **Moving-block bootstrap** 95% CI for r (block length $\lceil n^{1/3} \rceil$, 5000 resamples of the *paired* series, preserving within-block memory).
- **Block-permutation** null p : reorder contiguous blocks of x against a fixed y , destroying the cross-correlation while approximately preserving each series’ autocorrelation (5000 draws).
- **Multiple testing**: a Benjamini–Hochberg false-discovery-rate (FDR) correction (q -values) across the entire scale $\times$ variant grid. A cell is *robust* only if $q_{\text{block}} < 0.05$ and the bootstrap CI excludes zero.

The pipeline is reproducible (`analysis/nox008_spatial_scale.py`, seed 20260614); it reads only the committed benchmark and the gitignored NO₂ cube.

Table 1: Full scale $\times$ variant grid (monthly; footprint–background NO₂, meteo-normalised, vs the CREA BF operating rate). r , p , n_{eff} and the moving-block bootstrap 95% CI are at lag 0; q is the Benjamini–Hochberg FDR-adjusted block-permutation p across all 16 cells; the last column is the peak- $|r|$ lead-lag r (months). *No cell is naive-significant; none is robust.*

Scale	Variant	n	r_0	p	n_{eff}	boot 95% CI	q	best lag r
wide-native	level	90	-0.049	0.648	66.7	$[-0.31, +0.16]$	0.99	-0.31 (-5)
wide-native	intensity	90	+0.126	0.236	71.9	$[-0.12, +0.30]$	0.99	+0.26 (+6)
wide-native	yoy	78	-0.003	0.976	55.8	$[-0.40, +0.29]$	0.99	-0.26 (-6)
wide-native	stl	90	-0.002	0.984	84.0	$[-0.22, +0.17]$	0.99	-0.23 (-5)
wide-coarse010	level	90	+0.186	0.079	62.4	$[-0.07, +0.36]$	0.99	+0.28 (+4)
wide-coarse010	intensity	90	+0.131	0.217	62.9	$[-0.13, +0.31]$	0.99	+0.23 (+4)
wide-coarse010	yoy	78	-0.065	0.572	54.3	$[-0.52, +0.24]$	0.99	-0.30 (-6)
wide-coarse010	stl	90	+0.108	0.312	77.0	$[-0.11, +0.27]$	0.99	+0.24 (+6)
tight-native	level	90	-0.081	0.447	66.2	$[-0.34, +0.13]$	0.99	-0.34 (-5)
tight-native	intensity	90	+0.093	0.383	71.4	$[-0.14, +0.26]$	0.99	+0.27 (+6)
tight-native	yoy	78	-0.031	0.790	55.3	$[-0.41, +0.25]$	0.99	-0.28 (-6)
tight-native	stl	90	+0.019	0.859	85.6	$[-0.21, +0.20]$	0.99	+0.22 (+6)
tight-coarse010	level	90	-0.007	0.948	69.1	$[-0.27, +0.19]$	0.99	-0.25 (-5)
tight-coarse010	intensity	90	+0.048	0.653	69.6	$[-0.21, +0.24]$	0.99	+0.26 (+6)
tight-coarse010	yoy	78	-0.119	0.300	56.0	$[-0.49, +0.19]$	0.99	-0.31 (-7)
tight-coarse010	stl	90	-0.044	0.682	77.6	$[-0.31, +0.15]$	0.99	-0.21 (-5)

4 Results

4.1 No cell is significant at any scale

Table 1 is the full grid. Of the 16 scale $\times$ variant cells, the largest lag-0 $|r|$ is 0.19 (wide-coarse010, level); **none** reaches even the naive $p < 0.05$ threshold, and after the FDR correction every $q \approx 0.99$. The bootstrap CIs all straddle zero. The peak- $|r|$ lead-lag columns reach $|r| \approx 0.30$ –0.34, but those are selected over 17 candidate lags and are exactly what the block-permutation null is built to deflate: not one survives. Figure 1 renders the lag-0 grid; the palette barely departs from white.

4.2 Serial dependence inflates the naive test

Figure 2 plots the autocorrelation-robust block-permutation p against the naive i.i.d. p for every cell. Points sit on or above the diagonal: the honest p is never smaller than the naive one. The effective sample size (Table 1) shrinks the nominal $n = 90$ to $n_{\text{eff}} \approx 62$ –86 — a 5–31% loss of independent information — and the yoy variant, which differences the series, loses the most. In this dataset the naive test happens to agree (nothing is significant either way), but the gap is the general warning: a single fixed-scale Pearson p near 0.05 on these series would be a false positive.

4.3 The levels coupling is weak and sign-unstable

Figure 3a is a forest plot of the bootstrap CIs for the level variant across scales. Every interval crosses zero, and the point estimates change sign with scale ($-0.05, +0.19, -0.08, -0.01$). The negative “decoupling” levels relationship that read ~ -0.35 on the partial cube is, on the complete cube, a near-zero estimate whose sign is not even stable to the AOI buffer or the grid factor. Figure 3b shows why directly: the BF rate climbs steadily while the meteo-normalised cluster NO₂ hovers around zero with no matching trend — visual decoupling, $r = -0.05$, $n = 90$.

NO₂(footprint-bg) vs BF rate — lag-0 r across spatial scales (monthly)

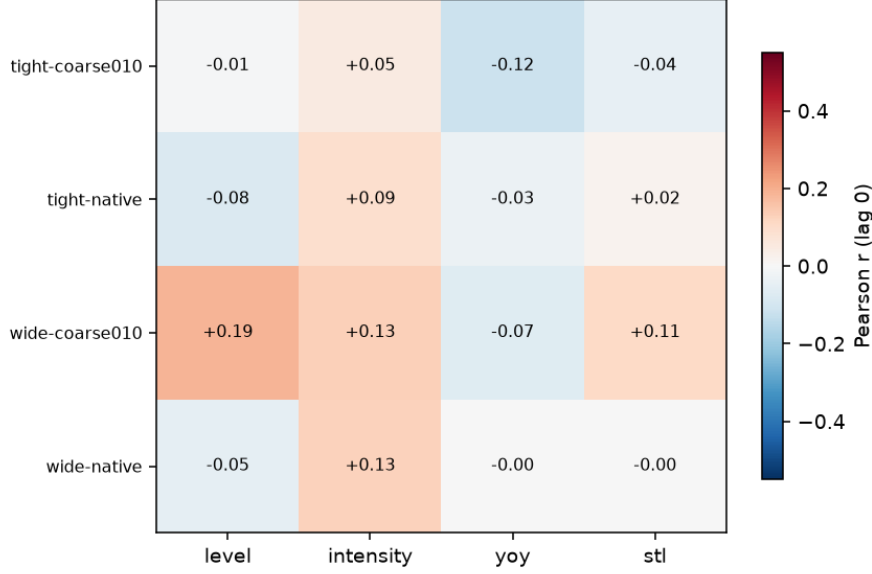


Figure 1: Lag-0 Pearson r between the meteo-normalised footprint–background NO₂ and the BF operating rate across the four resolvable scales (rows) and four deseason variants (columns). All $|r| < 0.2$; the sign of the level relationship even flips between *wide-native* (-0.05) and *wide-coarse010* ($+0.19$).

4.4 Regime dependence, not a stable signal

The only place the coupling is ever strong is *locally in time*. Figure 4 shows the 18-month rolling correlation of the *level* and *intensity* variants against the BF rate at the default scale: it wanders between strongly positive and strongly negative without settling. A rolling window will always surface a transiently high $|r|$ in a pair of autocorrelated series; that is a property of the estimator, not evidence of a usable signal, and it is precisely why the full-window, scale-swept, FDR-corrected test in Table 1 is the appropriate arbiter.

4.5 The coarsest grid cannot resolve a source contrast

At $\sim 0.25^\circ$ (~ 25 km cells) the footprint/background geometry collapses: within the clipped AOI the background ring (inner ~ 71 km, outer ~ 106 km once floored to the coarse cell size) contains no cells, for both the wide and the tight extent (**GeometryError**, ERR-003). This is itself the Parubets and Naito (2025) warning in its strongest form: at a coarse enough resolution the cluster and its background are no longer separable objects, so the source-isolation signal is undefined, not merely weak. The two *coarse025* scales are therefore reported as *not resolvable* rather than forced, and the grid in Table 1 comprises the four scales that admit a valid contrast.

5 Discussion

The null is robust to both suspicions. Neither a tighter AOI (less BTH background dilution) nor a coarser grid (more spatial aggregation) rescues the coupling, and the autocorrelation-aware test — which can only *reduce* significance — does not promote any cell. The weak/null NO₂↔BF relationship is not a fixed-scale artefact and not an i.i.d.-inflation artefact. The earlier -0.35 “decoupling” was a feature of the partial cube; the complete 389-week series collapses it to -0.05 , consistent with the documented full-series arbiter (weekly 0.07, monthly 0.13).

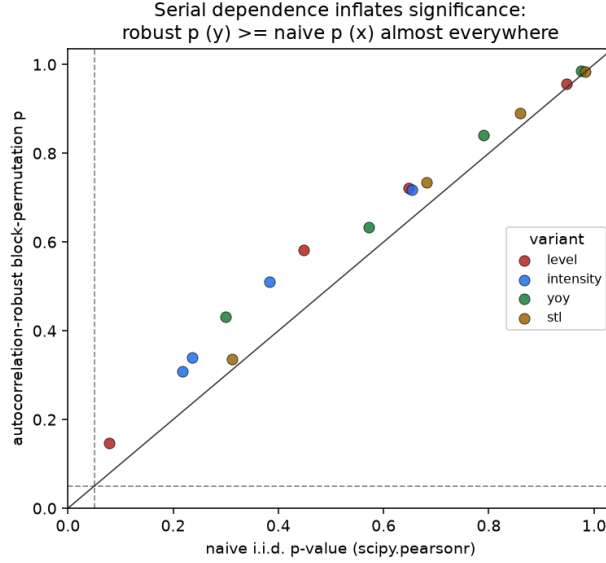


Figure 2: Autocorrelation-robust p (block permutation) vs the naive i.i.d. p (`scipy.pearsonr`) for all 16 cells. Every point lies on or above the 45° line: serial dependence can only *inflate* significance, never deflate it. Dashed lines mark $p = 0.05$.

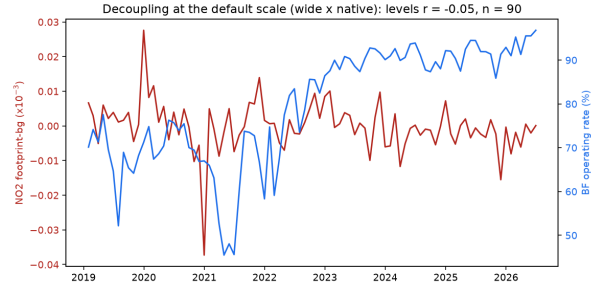
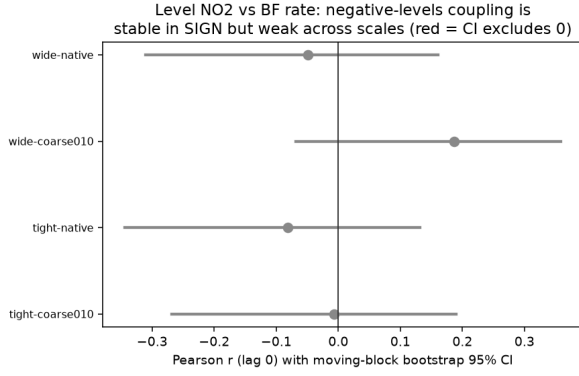
Why a level estimator is capped here. Steel is only $\sim 30\text{--}43\%$ of the Tangshan NO_2 column (Sci. Rep., 2024); traffic, power and residential sources fill the rest, and the ultra-low-emission retrofit programme drives a secular intensity decline that decouples emissions from output (Li et al., 2024). With steel a minority, season-dependent share of a noisy column (Montgomery and Holloway, 2018), a level-correlation estimator faces a signal-to-noise ceiling that more spatial slicing of a single daily overpass cannot lift. This is the same ceiling NOX-006 hit from the temporal side: TROPOMI’s lone $\sim 13:30$ sample cannot separate baseload steel from the cyclical traffic that would otherwise be removed by a day-of-week term.

Methodological takeaway. The experiment also demonstrates a discipline worth keeping: sweeping scale *and* lag multiplies the number of tests, and any single “significant” cell must be read against that multiplicity. The combination of a moving-block bootstrap, a block-permutation null, and a Benjamini–Hochberg correction (Morris and Zhang, 2019) converts a tempting collection of $|r| \sim 0.3$ peak-lag values into the honest conclusion they deserve.

What could still move the needle. Two levers remain, both outside this paper’s aggregation-only scope. (i) A *diurnal* sensor: GEMS, the geostationary spectrometer over East Asia, samples ~ 8 times per day and could resolve the weekday cycle that separates flat steel baseload from cyclical traffic, lifting steel’s *variance* share above the mean-share ceiling (Park et al., 2025). (ii) *Flux-divergence* point-source attribution (Beirle et al., 2021, 2023), which isolates the local source term from the regional background physically rather than statistically — known to be hard in high-background, complex-wind industrial NE China, which is exactly why it is sequenced after a relative-index null is established. NOX-008’s robust null is the go-signal that a relative-index, single-overpass approach has been exhausted at this cluster.

6 Conclusion

On the complete TROPOMI cube, the Tangshan steel-activity NO_2 signal shows no robust correlation with the blast-furnace operating rate at *any* reasonable spatial scale, under an autocorrelation-



(a) Bootstrap 95% CI of r (level) across scales; all cross zero. (b) Default scale: BF rate (blue) trends up; NO_2 (red) is flat.

Figure 3: The levels relationship at and across the default scale. Left: the moving-block bootstrap CI of the lag-0 r for the level variant at every resolvable scale — each interval includes zero, and the central estimate changes sign. Right: the monthly time series at the default (wide $\times$ native) scale.

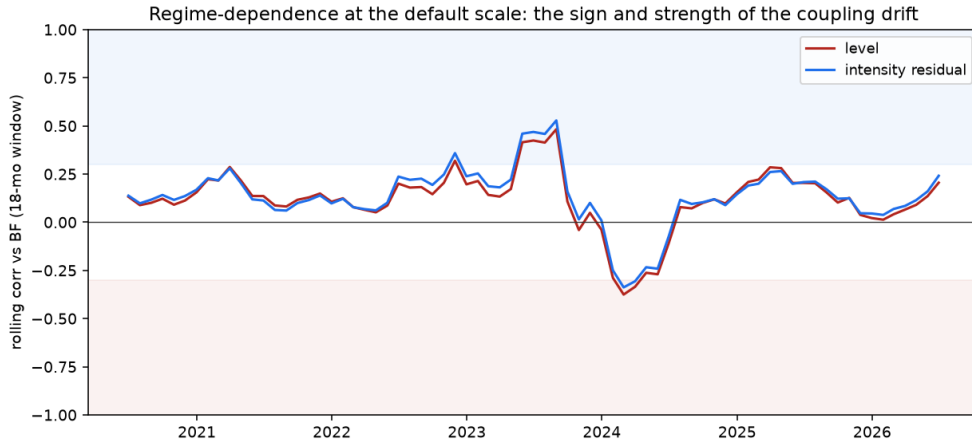


Figure 4: 18-month rolling correlation vs the BF rate at the default scale for the level (red) and intensity-residual (blue) variants. The sign and strength drift across regimes; no stable coupling emerges. Shaded bands mark $|r| \geq 0.3$.

aware, multiple-testing-corrected test: 0/16 cells significant, sign-unstable in levels, and undefined at the coarsest grid. The result is not a failure of the pipeline but a sharpened, defensible boundary on what a single-overpass relative-index method can deliver for one industrial cluster — and a clean motivation for the diurnal-sensor and flux-divergence phases that follow.

Reproducibility. All figures and Table 1 are produced by `analysis/nox008_spatial_scale.py` (seed 20260614, 5000 bootstrap/permutation draws), reading `data/derived/no2/no2_cube_w.nc` (gitignored) and `data/derived/benchmark_tangshan_bf_operating_rate.parquet`. Outputs: `docs/figures/nox008/`. NO_2 -derived artefacts are gitignored; the stats table and figures are committable.

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